

Course Syllabus

 Edit

Instructor

Name: Dr. David Patrick

Email: ueg11@txstate.edu (<mailto:ueg11@txstate.edu>)

Include CS 1428 in the subject head as it allows me to respond to these emails faster.

*I typically respond within 24 hours on weekdays.

Office Hours

Hours: MW 12:30 - 2:00 and by appointment

Location: CMAL 210E

If my office door is open (and it isn't within 15 minutes of a class time), feel free to enter.

Teaching Assistants

Name: Kamrul Hasan

Email: sii25@txstate.edu (<mailto:sii25@txstate.edu>)

Name: Awatif Yasmin

Email: nuc4@txstate.edu (<mailto:nuc4@txstate.edu>)

*Please allow several hours for a response and do not expect responses over weekends, holidays, or at night.

Course Format

This is a **Face to Face** course.

Section 001 Class Hours: MWF 9:00 am - 9:50am

Section 002 Class Hours: MWF 10:00 am - 10:50am

Location: Bruce and Gloria Ingram Hall 03104

Textbook

We will be using an online textbook from zyBooks. It is **required**. Please visit Module 0 on Canvas for how to get access to the Zybook.

Course Description

Introductory course for computer science majors, minors and others desiring a technical introduction to computer science. The course emphasizes problem solving, algorithm development, structured programming, good coding style, and programming in C++.

Course Prerequisites

MATH 1315 or equivalent

Course Goals and Learning Objectives

At the end of the course, the students should be able to:

- Describe the properties of good algorithms.
- Design and develop good algorithms using a top-down approach.
- Use the C++ programming language to implement, test, and debug algorithms for solving simple problems.
- Explain the concepts of data types, variables, and literals and use them in programs.
- Write C++ code that solves computational problems.
- Use an if or if-else construct to implement branching in an algorithm.
- Use a for loop for definite iteration.
- Use a while or do-while loop for indefinite iteration.

- Use functions and parameters to simplify longer programs and reuse code from previous solutions.
- Demonstrate the mechanics of parameter passing with emphasis on the difference between pass by value and pass by reference.
- Manipulate data in arrays.
- Create a new data type by using a structure.
- Analyze and explain the behavior of simple programs involving the fundamental programming constructs covered in this class.
- Modify and expand short programs that use the constructs covered in this class.
- Describe strategies that are useful in debugging.
- Use a Windows- or Mac-based editor and compiler environment to develop programs in C++.

Lab Section

This course includes a separate lab section taught by a lab instructor. The lab reinforces concepts covered in lecture through hands-on activities and additional practice. Lab attendance and participation are required. For specific lab policies, schedule, and instructor information, refer to your lab syllabus or contact your lab instructor directly.

Required Material

- ZyBooks

Zybooks

This interactive textbook contains readings, live coding examples from class, and most course assignments. Components typically include Learning Activities, Take Home Quizzes (THQs), and Large Scale Programs (LSPs). Given the substantial number of activities throughout the semester, it is strongly recommended that you pace yourself and complete them consistently rather than attempting to finish everything at the last minute.

Grading

The course is graded out of an 1000 point scale. Below is a breakdown of the course's point structure:

Course Point Structure

<u>Group</u>	<u>Points</u>
ZyBook Learning Activities	60
Take Home Quizzes (THQs)	105
Large Scale Programs (LSPs)	265
Lab Section	100
Midterm Exam	150
Final Exam	300

1. Rounding Policy:

- Final grades are determined based on the total points earned out of 1000. Fractional points will not automatically be rounded up.
- **Failing Exam Policy:** If a student fails either exam (scores below 60%), they may not qualify for any rounding considerations, regardless of their total points.
- In cases of borderline scores (e.g., 899 points), adjustments may be made at the instructor's discretion, particularly if the student's overall performance reflects consistent effort and improvement. However, this is not guaranteed.

2. Grading Scale:

- **A:** 900+ points
- **B:** 800–899 points
- **C:** 700–799 points
- **D:** 600–699 points
- **F:** 0–599 points

3. Extra Credit Policy:

- Extra points may be made available throughout the semester through bonus activities, challenges, or participation incentives. These points can serve as "padding" to help students recover from lost points elsewhere since the grading scale remains fixed at 1000 points.

4. Flexibility of Point Structure:

- The assignment groups and their totals outlined in this syllabus are subject to change. Adjustments may be made to accommodate unforeseen circumstances, to improve the course structure, or to better reflect the workload and learning outcomes. Any changes to

the assignment points or grading policy will be communicated through Canvas and during class.

Program Submission and Grading Policy

All submitted programs must compile and run to be eligible for grading. Programs that fail to compile or do not execute correctly will receive a grade of zero. It is your responsibility to thoroughly test your code to ensure it functions as expected. Partial credit will not be awarded for non-functioning programs.

Makeup/Late Policy

Assignments and quizzes cannot be turned in late or made up without prior permission from the instructor.

Exam Policy

- All tests will be in class during class time.
- There will be no makeup tests. If you have an instructor-approved excuse for missing the midterm, your final exam score will be used as the score for both exams. This does not increase the weight of the final; it simply substitutes the missing midterm score.

Attendance Policy

Class attendance is highly encouraged but it is not directly included as part of your final grade. Programming is a skills-based activity, and missing class means missing practice. Falling behind is hard to recover from.

Student Disability Accommodations

Any student with needs requiring special accommodations should contact the office of disability

services (ods.txst.edu → <http://ods.txst.edu>). Students who qualify for extra time for exams must take their test with ATSD and must schedule their test at the same time the test is given in class. You must submit your request with ATSD **at least 3 business days before the exam!**

Helpful Resources

Other than the instructor's office hours, there are other places to obtain assistance:

- **CS Lab tutors** (good for help on MSPs): MCS 593 (inside MCS 590 suite)
<https://cs.txstate.edu/resources/labs/tutoring/> (<https://cs.txstate.edu/resources/labs/tutoring/>)
- **CLC tutoring** (good for help on understanding the material):
<https://hlsamp.cose.txstate.edu/clc.html> (<https://hlsamp.cose.txstate.edu/clc.html>)
- **TA's office hours**: see Canvas for times and how to access.

Academic Honesty

You are expected to adhere to the University's Academic Honor Code: <https://www.txstate.edu/honorcodecouncil/Academic-Integrity.html> (<https://www.txstate.edu/honorcodecouncil/Academic-Integrity.html>)

- You may discuss problems with other students but do not share your code electronically with anyone. Do not post your code on group chats or in email!
- Do not include code obtained from the internet or any other source in your programs. Do not use features we have not covered in class without prior approval of the instructor.
- Do no collaborate with anyone during a test or exam (no talking, no cell phones or smart watches).
- The penalty for violating academic honesty on an assignment, test or exam is a reduction of points up to a **negative grade** of the assignment, test or exam.

Artificial Intelligence (AI) Use in This Course

What Counts as AI

AI refers to any program or tool that can generate or edit content, including ChatGPT, Copilot, and AI features in development environments.

How You Can Use AI

You may use AI tools to support your learning, such as:

- Brainstorming
- Getting hints or asking clarifying questions
- Generating practice materials
- Reviewing explanations of course material
- Getting help understanding error messages

Think of AI as a study partner. It's helpful for practice and review, but not a replacement for your own work.

What's Not Allowed

- No AI-generated code: All code you turn in must be written and tested by you. Copying, adapting, or "tweaking" AI-generated code counts as prohibited use.
- No AI-submitted work: You cannot submit AI-generated writing or explanations as your own.
- No auto-complete coding from AI: If your IDE/editor has code generation features (e.g., Copilot), you need to turn them off or actively ignore them. You're responsible for ensuring your work is your own.

Why These Restrictions Exist

This course is about developing your own programming and problem-solving skills. If you rely on AI to write your code, you skip the practice you need to become a strong programmer. By writing your own solutions, you'll build skills that AI tools can't give you.

Your Responsibility

If you use AI in permitted ways, you must:

- Make sure you fully understand the explanations it provides
- Write your own original code from scratch
- Be able to explain and reproduce your solutions if asked

Misusing AI not only undermines your learning, but may also violate academic integrity policies. The goal is for you to leave this class confident in your own ability to solve problems and write programs independently.

Withdrawals/Drops

You must follow the withdrawal and drop policy set up by the University.

<https://www.registrar.txstate.edu/resources/dropping-vs-withdrawing.html> (<https://www.registrar.txstate.edu/resources/dropping-vs-withdrawing.html>)

Course Schedule

Below is the proposed course schedule:

Course Schedule

<u>Week</u>	<u>Topic</u>
0	Intro to Programming and C++
1	Intro to Programming and C++
2	Expressions and Data Types
3	Expressions and Data Types
4	Branches
5	Branches
6	Loops
7	Midterm Review
8	Loops
9	Functions
10	Functions
11	Arrays
12	Arrays
13	Thanksgiving Break Holiday (Optional Lecture)
14	Final Review

Minor changes to the course schedule may occur throughout the semester to better align with the course content and objectives. All updates and adjustments will be reflected on the live calendar located on the course homepage. It is your responsibility to regularly check the live calendar for the most current information. Any significant changes will also be announced in class and via Canvas announcements.

For a list of important university-wide dates, review [Office of the University Registrar Academic Calendar](https://www.registrar.txst.edu/registration/ac/academic-calendar.html)  (<https://www.registrar.txst.edu/registration/ac/academic-calendar.html>)

Exam Topics

Midterm will cover Units 1, 2, and 3.

Final exam will be a comprehensive exam of Units 1-6.